Exact time limit for first-order detuning-robust qubit π gates

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We establish the exact minimum duration of a qubit π gate that cancels a constant detuning error to first order. For arbitrary measurable transverse controls satisfying $u_x^2 + u_y^2 \le \Omega^2$, the minimum is $7\pi/(3\Omega)$. The only controls attaining it are the known short-CORPSE sequence and its overall negative, up to changes on sets of measure zero. A second independently controlled transverse axis therefore provides no time advantage for this task. The lower bound follows from a sharp inequality valid in every dimension at least two: a unit-vector curve with speed at most one, antipodal endpoints, and zero time average has duration at least $7\pi/3$. Projection onto the curve's value at the time midpoint gives the comparison and determines its equality cases. The argument applies directly to measurable controls without requiring a classification of extremal trajectories.

I. INTRODUCTION

A finite drive amplitude sets a time cost for suppressing systematic errors during a quantum gate. Independent amplitude and phase control enlarge the available pulse shapes, raising the question of whether this freedom can reduce that cost. We address this question for a full qubit π gate with first-order robustness against static detuning, under a joint amplitude bound on two transverse controls.

The short-CORPSE sequence has duration $7\pi/(3\Omega)$ and is optimal within a symmetric family of three elementary pulses [1]. Related planar geometric constructions were developed for a single control axis [2]. With two independent transverse controls, the time-optimality of short-CORPSE has remained a separate question: Pontryagin analysis identifies it as a candidate [3], and numerical optimization gives first-order robust X-gate durations near $2.33\pi/\Omega$ [4].

The control problem is closely related to bounded-curvature paths. The planar three-arc return with reversed tangent appears as a teardrop trajectory in the controllability construction of Sigalotti and Chitour [5, Remark 2.2]. Kim and Cheong explicitly give $7\pi/3$ as the shortest planar bounded-curvature path between coincident positions with opposite directions [6, Fig. 7 and the case $d = 0$]. The corresponding spatial problem permits additional trajectories: general three-dimensional shortest-path and reachability analyses include nonplanar extremals [7–9]; related spatial branches also occur in the classification of ∞-elastica [10].

Here we prove that short-CORPSE remains time optimal over the full class of measurable two-axis controls. The key step is a comparison inequality for spatial curves, which supplies the lower bound beyond the planar problem. This Schur-type tangent comparison [11] projects onto the midpoint tangent and uses the antipodal endpoints to constrain both halves of the curve. Equality forces the curve into a single plane and, with the full gate condition, fixes the control to short-CORPSE or its overall negative.

II. CONTROL MODEL AND MAIN RESULT

We set $\hbar = 1$ and take Pauli matrices with eigenvalues ± 1. The propagator obeys

$$\begin{aligned} \dot{U}_\delta(t) &= -\frac{i}{2}\big[u_x(t)\sigma_x + u_y(t)\sigma_y + \delta\sigma_z\big]U_\delta(t), \\ U_\delta(0) &= I. \end{aligned} \tag{1}$$

Here δ is an unknown constant and the real controls are measurable, independent of δ, and satisfy the amplitude bound

$$u_x(t)^2 + u_y(t)^2 \le \Omega^2 \quad \text{for almost every } t, \tag{2}$$

where $\Omega > 0$ is fixed. There is no independent longitudinal control. A change of control phase changes the subsequent transverse drive; it does not apply an additional instantaneous gate. The admissible class includes reduced amplitudes, zero-control intervals, and arbitrary phase changes.

Write $R_x(\theta) = \exp(-i\theta\sigma_x/2)$. Let T_* be the infimum of all durations $T > 0$ for which an admissible control satisfies

$$\begin{aligned} U_0(T) &= \pm R_x(\pi), \\ \partial_\delta U_\delta(T)|_{\delta=0} &= 0. \end{aligned} \tag{3}$$

The sign ignores an overall phase. The derivative condition is an operator identity, so robustness holds for every input state.

Theorem 1 (Exact robust gate time)**.** *For the control class* (2) *and target* (3)*,*

$$\boxed{T_* = \frac{7\pi}{3\Omega}.} \tag{4}$$

Apart from changes on sets of measure zero, the only minimizing controls have $u_y = 0$ and $u_x = \pm\Omega q(\Omega t)$, where q has three successive constant segments of values $+1, -1, +1$ and dimensionless durations $\pi/3, 5\pi/3, \pi/3$.

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Thus the second transverse axis does not shorten this robust gate. The lower bound applies to the full measurable control class; planarity and the three-segment structure follow from equality.

A. The toggling-frame error curve

To obtain the lower bound, introduce the toggling-frame noise direction and its integral:

$$\begin{aligned} \mathbf{n}(t)\cdot\boldsymbol{\sigma} &= U_0(t)^\dagger \sigma_z U_0(t), \\ \mathbf{r}(t) &= \int_0^t \mathbf{n}(s)\,\mathrm{d}s. \end{aligned} \tag{5}$$

This is the error-curve representation used in geometric robust control [2, 12]. The identities needed here follow directly from (1). Differentiation with respect to the constant parameter δ gives the variation-of-constants formula

$$\begin{aligned} \partial_\delta U_\delta(T)|_0 &= -\frac{i}{2} U_0(T) \int_0^T U_0(t)^\dagger \sigma_z U_0(t)\,\mathrm{d}t \\ &= -\frac{i}{2} U_0(T)\,\mathbf{r}(T)\cdot\boldsymbol{\sigma}. \end{aligned} \tag{6}$$

Hence first-order detuning cancellation is equivalent to $\mathbf{r}(T)=0$.

Unitary conjugation preserves the Pauli inner product, so $|\mathbf{n}|=1$. Define the orthonormal vectors $\mathbf{a},\mathbf{b},\mathbf{n}$ by conjugating $\sigma_x,\sigma_y,\sigma_z$, respectively. The commutator in the derivative of (5) yields

$$\begin{aligned} \dot{\mathbf{n}} &= u_x\mathbf{b} - u_y\mathbf{a}, \\ |\dot{\mathbf{n}}| &= \sqrt{u_x^2+u_y^2} \le \Omega \quad \text{a.e.} \end{aligned} \tag{7}$$

The endpoint gate imposes

$$\mathbf{n}(0)=\mathbf{e}_z, \qquad \mathbf{n}(T)=-\mathbf{e}_z, \tag{8}$$

because $R_x(\pi)^\dagger\sigma_z R_x(\pi) = -\sigma_z$. For bounded measurable controls, U_0 and $\mathbf{n}$ are Lipschitz and $\mathbf{r}\in W^{2,\infty}$; all differential identities above hold almost everywhere.

Rescaling time by $s=\Omega t$ thus produces an absolutely continuous unit-vector curve of duration $L=\Omega T$, speed at most one, antipodal endpoints, and zero integral. The following theorem proves $L\ge 7\pi/3$. The argument uses this forward map from controls to curves; a direct calculation in Sec. IV establishes attainment in the gate model.

III. A SHARP MIDPOINT COMPARISON

Theorem 2 (Antipodal zero-mean spherical curves)**.** *Let $d\ge 2$ and let $\mathbf{n}:[0,L]\to S^{d-1}$ be absolutely continuous, where $L>0$. Suppose that*

$$\begin{aligned} |\mathbf{n}'(t)| \le 1 \quad \textit{a.e.}, \qquad \mathbf{n}(L) &= -\mathbf{n}(0), \\ \int_0^L \mathbf{n}(t)\,\mathrm{d}t &= 0. \end{aligned} \tag{9}$$

Then $L\ge 7\pi/3$. The constant is sharp.

Proof. Set $c=L/2$ and $\mathbf{v}=\mathbf{n}(c)$. Spherical distance is bounded by curve length, so

$$d_S(\mathbf{n}(s),\mathbf{n}(t)) \le |t-s|. \tag{10}$$

If $c<\pi$, this gives

$$\begin{aligned} \mathbf{v}\cdot\int_0^{2c}\mathbf{n}(t)\,\mathrm{d}t &\ge \int_0^{2c}\cos|t-c|\,\mathrm{d}t \\ &= 2\sin c > 0, \end{aligned} \tag{11}$$

contrary to the zero integral. We may therefore assume $c\ge\pi$.

Let $\alpha = d_S(\mathbf{v},\mathbf{n}(0))$. The antipodal endpoint has distance $\pi-\alpha$ from $\mathbf{v}$. Consider either half of the curve, oriented from its endpoint towards $\mathbf{v}$, and denote it by $\mathbf{m}:[0,c]\to S^{d-1}$. If $\beta = d_S(\mathbf{v},\mathbf{m}(0))$, the triangle inequality and (10) imply

$$d_S(\mathbf{v},\mathbf{m}(t)) \le \min\{\beta+t, c-t, \pi\}. \tag{12}$$

Since cosine is decreasing on $[0,\pi]$,

$$\int_0^c \mathbf{v}\cdot\mathbf{m}(t)\,\mathrm{d}t \ge A(c,\beta), \tag{13}$$

where direct integration gives

$$\begin{aligned} A(c,\beta) &:= \int_0^c \cos\big(\min\{\beta+t, c-t, \pi\}\big)\,\mathrm{d}t \\ &= \begin{cases} 2\sin\big((c+\beta)/2\big) - \sin\beta, & c+\beta\le 2\pi, \\ 2\pi - c - \beta - \sin\beta, & c+\beta\ge 2\pi. \end{cases} \end{aligned} \tag{14}$$

Indeed, in the first case the two affine bounds meet at $t=(c-\beta)/2$. In the second, their minimum is capped at π for an interval of length $c+\beta-2\pi$. The formula applies because $0\le\beta\le\pi\le c$. Applied to both halves, (13) yields

$$0 \ge A(c,\alpha) + A(c,\pi-\alpha). \tag{15}$$

We evaluate the right-hand side at $c_* = 7\pi/6$. Write

$$B(\alpha) = A(c_*,\alpha) + A(c_*,\pi-\alpha). \tag{16}$$

By symmetry, it suffices to consider $0\le\alpha\le\pi/2$. If $\pi/6\le\alpha\le\pi/2$, both terms use the first branch of (14). With $x=\cos(\alpha/2-\pi/4)\in[\sqrt{3}/2,1]$, one obtains

$$\begin{aligned} B(\alpha) &= 4\sin(5\pi/6)x - 2\sin\alpha \\ &= 2x - 2(2x^2-1) = 2(1-x)(2x+1) \ge 0. \end{aligned} \tag{17}$$

If $0\le\alpha\le\pi/6$, the second term in (16) uses the capped branch, so

$$\begin{aligned} B(\alpha) &= 2\sin(7\pi/12+\alpha/2) \\ &\quad + \alpha - \pi/6 - 2\sin\alpha. \end{aligned} \tag{18}$$

Its derivative is strictly negative on this interval:

$$B'(\alpha) = \cos(7\pi/12 + \alpha/2) + 1 - 2\cos\alpha < 0. \tag{19}$$

Consequently $B(\alpha) \geq B(\pi/6) = \sqrt{3} - 1 > 0$. Together with symmetry, these calculations show

$$B(\alpha) \geq 0 \quad (0 \leq \alpha \leq \pi), \qquad B(\alpha) = 0 \iff \alpha = \pi/2. \tag{20}$$

Finally, fix $\beta \in [0, \pi]$. For $\pi < c \leq c_*$, the derivative of A with respect to c is either $\cos((c+\beta)/2) < 0$ in the uncapped branch or -1 in the capped branch. These derivatives agree at the branch boundary. Thus, whenever $\pi \leq c < c_*$,

$$A(c, \beta) > A(c_*, \beta). \tag{21}$$

Equations (15), (20), and (21) would then give

$$0 \geq A(c, \alpha) + A(c, \pi - \alpha) > B(\alpha) \geq 0, \tag{22}$$

a contradiction. Hence $c \geq c_*$, or $L \geq 7\pi/3$. The planar curve constructed in Sec. IV attains equality and embeds into every S^{d-1} with $d \geq 2$. □

The same result can be stated directly for spatial curves. If $\mathbf{r} \in W^{2,\infty}([0, L]; \mathbb{R}^d)$ has $|\mathbf{r}'| = 1$, $|\mathbf{r}''| \leq 1$ almost everywhere, $\mathbf{r}(L) = \mathbf{r}(0)$, and $\mathbf{r}'(L) = -\mathbf{r}'(0)$, then its length is at least $7\pi/3$. Taking $\mathbf{n} = \mathbf{r}'$ proves this assertion. Self-intersections and curvature discontinuities are allowed.

IV. ATTAINMENT AND ALL EQUALITY CASES

A. The attaining pulse

For completeness, we verify the known attaining pulse in the same normalization as the lower bound; see Fig. 1. Use dimensionless time $s = \Omega t$ and set $u_y = 0$ and

$$\frac{u_x}{\Omega} = q(s) := \begin{cases} +1, & 0 \leq s < \pi/3, \\ -1, & \pi/3 \leq s < 2\pi, \\ +1, & 2\pi \leq s \leq 7\pi/3. \end{cases} \tag{23}$$

Let $\phi(s)$ be the accumulated signed rotation angle. At the three successive segment endpoints it takes the values $\pi/3$, $-4\pi/3$, and $-\pi$. Since all three Hamiltonians commute,

$$U_0(T) = R_x(-\pi) = -R_x(\pi). \tag{24}$$

Moreover,

$$\mathbf{n}(s) = (0, \sin\phi(s), \cos\phi(s)). \tag{25}$$

The x component of its integral vanishes. For the other two, integration on the three segments gives

$$i\int_0^{7\pi/3} e^{i\phi(s)}\,\mathrm{d}s = 2e^{i\pi/3} - 2e^{-i4\pi/3} + e^{-i\pi} - 1 = 0. \tag{26}$$

The real and imaginary parts therefore establish $\mathbf{r}(T) = 0$, and (6) proves first-order gate robustness. This supplies the upper bound matching Theorem 2.

The pulse also explains sharpness of the midpoint comparison. At $c_* = 7\pi/6$ its accumulated angle is $-\pi/2$, so the midpoint tangent is $\mathbf{v} = -\mathbf{e}_y$, perpendicular to both endpoint tangents. On either half of the curve, oriented from its endpoint towards the midpoint, the spherical distance to $\mathbf{v}$ is exactly

$$d_S(\mathbf{v}, \mathbf{m}(t)) = \min\{\pi/2 + t, c_* - t\}, \quad 0 \leq t \leq c_*. \tag{27}$$

Its projected integral is $A(c_*, \pi/2) = 2\sin(5\pi/6) - 1 = 0$ on each half. Thus the pulse attains both the pointwise distance comparison and its integrated zero bound.

B. Geometric rigidity and uniqueness of the control

The comparison also determines all minimizers. Suppose that $L = 7\pi/3$. Equation (20) forces $\alpha = \pi/2$. The two continuous, nonnegative gaps in (13) must then vanish identically. Orient each half from the midpoint towards its endpoint, calling it $\mathbf{m} : [0, c_*] \to S^{d-1}$. Its distance from $\mathbf{v}$ is therefore

$$f(s) = \min\{s, 5\pi/3 - s\}, \qquad 0 \leq s \leq c_*. \tag{28}$$

For $s > 0$, write

$$\mathbf{m}(s) = \cos f(s)\,\mathbf{v} + \sin f(s)\,\mathbf{e}(s), \tag{29}$$

where $\mathbf{e}(s) \perp \mathbf{v}$ and $|\mathbf{e}(s)| = 1$. Because $0 < f(s) \leq 5\pi/6 < \pi$, the vector $\mathbf{e}$ is absolutely continuous on every $[\varepsilon, c_*]$. Almost everywhere there,

$$|\mathbf{m}'|^2 = (f')^2 + \sin^2 f\,|\mathbf{e}'|^2 \leq 1. \tag{30}$$

As $|f'| = 1$ almost everywhere, $\mathbf{e}$ is constant. At the two endpoints $f(c_*) = \pi/2$, so these constants are respectively $\mathbf{n}(0)$ and $-\mathbf{n}(0)$. Thus both halves lie in the same great circle, and the only equality curves are the three-arc trajectory above, with arbitrary overall orientation.

For the gate problem this also fixes the controls. Given an equality curve with midpoint tangent $\mathbf{v} \perp \mathbf{e}_z$, let $\mathbf{a} = \mathbf{e}_z \times \mathbf{v}$. The explicit fixed-axis evolution $U_a(t) = \exp[-i\phi(\Omega t)\mathbf{a}\cdot\boldsymbol{\sigma}/2]$ generates that curve, where ϕ is the accumulated signed angle of pulse (23). If another admissible evolution U_0 generates the same $\mathbf{n}$, then $W = U_0 U_a^\dagger$ commutes with σ_z, hence is diagonal. Its derivative satisfies

$$\dot{W} = -iH_0 W + iWH_a, \tag{31}$$

where both nominal Hamiltonians H_0, H_a are off-diagonal in the σ_z basis. The right-hand side is off-diagonal, whereas $\dot{W}$ is diagonal. Therefore $\dot{W} = 0$ almost everywhere and $W = I$. Finally, $U_a(T) = i\mathbf{a}\cdot\boldsymbol{\sigma}$ equals $\pm R_x(\pi)$ only when $\mathbf{a} = \pm\mathbf{e}_x$. Up to changes on null

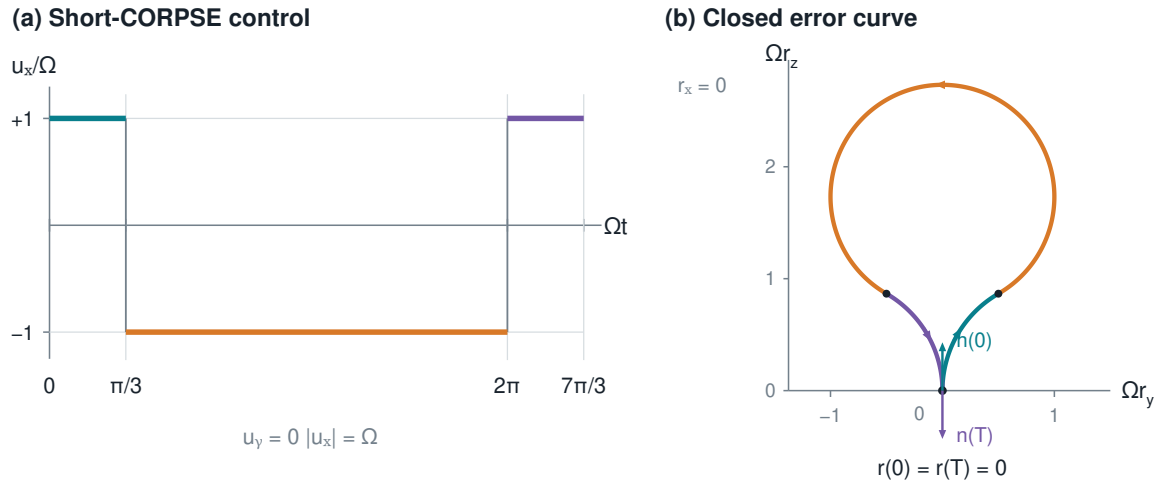


FIG. 1. Short-CORPSE attaining the robust π-gate time. (a) The saturated control segments have durations $(\pi/3, 5\pi/3, \pi/3)/\Omega$, with $u_y = 0$. (b) The planar error curve consists of three unit-radius circular arcs in dimensionless coordinates $\Omega\mathbf{r}$. It returns to its starting point with opposite initial and final tangents. Colors identify the same control segments in both panels.

sets, the only time-optimal controls are consequently (23) and its overall negative. This completes the uniqueness assertion in Theorem 1.

A Lean 4 formalization of the exact minimum time, attainment, and classification of all minimizing controls accompanies the source files.

V. DISCUSSION

The shortest error-free π gate has duration π/Ω: the endpoint condition (8) requires a spherical distance π, and a constant drive at amplitude Ω attains that time. First-order cancellation of static detuning increases the exact minimum by a factor 7/3. Geometrically, cancellation requires the integrated noise direction to return to the origin. Its turn rate is limited by the available drive, and the midpoint comparison quantifies the additional time required to balance that integral.

Allowing a second transverse component enlarges the set of possible noise-direction paths to spatial curves. Theorem 2 bounds this full class with the same constant achieved by a planar three-arc trajectory. This is why independent phase control gives no time advantage for the stated π-gate task. The proof includes arbitrary switching and nonsaturated amplitudes through the single almost-everywhere speed bound.

The spatial lower bound establishes the optimality of the known $7\pi/3$ planar construction [1, 5, 6] over the full control disk. Its equality conditions explain why the additional spatial freedom does not produce another minimizing control. The special antipodal endpoint identity is essential to the comparison. Other target angles carry different endpoint information and, for full gate synthesis, additional frame constraints. Determining their minimum robust times requires a separate analysis.

The convex planar reference-arc hypothesis in classical Schur comparison [11, Theorem 5.1] is not satisfied by this three-arc return. Moser's sufficient global-optimality criterion for ∞-elastica [10, Theorem 3], in the unweighted case, requires a nonzero auxiliary function g satisfying $0 \le g \le -\boldsymbol{\lambda}\cdot\mathbf{n}$ for a constant vector $\boldsymbol{\lambda}$. Zero total displacement would force $\int g = 0$, excluding that criterion here. The midpoint estimate supplies a comparison specifically adapted to antipodal tangents and zero displacement.

VI. CONCLUSION

Under the joint amplitude bound on arbitrary measurable transverse controls, the exact minimum time for a first-order detuning-robust qubit π gate is $7\pi/(3\Omega)$. Only short-CORPSE and its overall negative attain this time, with segment durations $(\pi/3, 5\pi/3, \pi/3)/\Omega$. The sharp midpoint comparison establishes both the time limit and the equality structure while preserving the full gate condition. Thus a second transverse control axis cannot reduce the time cost of first-order static-detuning cancellation for this gate.

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DECLARATIONS

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